

Topic:-

Prepare a chart giving detailed information of carbon compounds in everyday use. Display it in the class and discuss.

Introduction:-

Carbon is one of the most versatile and remarkable chemical elements in the entire universe. In the periodic table, carbon occupies atomic number 6 with electronic configuration 2, 4. Even though carbon makes up less than 0.03% of the Earth's crust and atmosphere, the total number of chemical compounds formed by carbon exceeds millions! In fact, an entire independent branch of science called **Organic Chemistry** is dedicated to the study of carbon compounds.

All living matter—our muscles, skin, blood, food nutrients (carbohydrates, proteins, fats), DNA genetic codes, medicines, fuels, soaps, perfumes, and synthetic fabrics—consists fundamentally of carbon compounds. Without carbon chemistry, life as we know it would simply not exist.

To explore the presence of carbon in our daily lives, our student team designed and prepared a comprehensive educational chart detailing major carbon compounds in everyday use, mounted the display in our classroom, and organized an interactive science discussion.

❖ Why Carbon Forms Millions of Compounds:-

- **Catenation:** Unique ability of carbon atoms to form strong covalent bonds with other carbon atoms, creating long straight chains, branched trees, and closed rings.
- **Tetravalency:** Having 4 valence electrons, carbon shares electrons with four atoms of hydrogen, oxygen, nitrogen, sulfur, or halogens.

Aim & Chart Preparation Plan:-

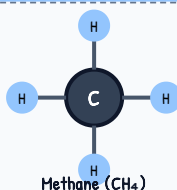
Aim: To design, prepare, and display an informative chart classifying everyday carbon compounds by their functional groups, molecular formulas, and daily uses, and lead a class seminar.

❖ Objectives of the Project:-

- To catalog everyday household items into organic chemical families (fuels, alcohols, acids, esters, polymers).
- To draw accurate structural formulas showing covalent bonds and functional groups.
- To discuss the cleansing mechanism of soaps and environmental aspects of synthetic plastics.

❖ Materials Required for Chart:-

- Large white drawing chart sheet (70 cm × 100 cm), colored sketch pens, marker pens, ruler.
- Class 10 Science & Technology Part 1 textbook (Topic: Carbon Compounds).
- Sample product wrappers: soap wrapper, vinegar bottle label, hand sanitizer label.



1. Hydrocarbons (LPG, CNG, Petrol)
2. Alcohols & Acids (Ethanol, Vinegar)
3. Polymers & Soaps (Polythene, Micelles)

Chemical Principles & Functional Groups:-

Carbon compounds are categorized into homologous series determined by functional groups:

- **1. Hydrocarbons (Alkanes, Alkenes, Alkynes):** Compounds composed exclusively of carbon and hydrogen. Saturated hydrocarbons (alkanes, single bonds C-C) are relatively unreactive and serve as primary fuels. Unsaturated hydrocarbons possess double (C=C) or triple (C≡C) bonds.
- **2. Alcohols (-OH Functional Group):** Contain the hydroxyl group attached to a carbon skeleton. Ethanol (C₂H₅OH) is the active component in medical spirit, sanitizers, and tinctures.
- **3. Carboxylic Acids (-COOH Functional Group):** Possess the carboxyl group. Ethanoic acid (CH₃COOH) is the chemical ingredient of household cooking vinegar.
- **4. Esters (-COO- Functional Group):** Produced by reacting a carboxylic acid with an alcohol. Esters have characteristic sweet fruity scents and are used in fruit essences and perfumes.

❖ Covalent Bonding Concept:-

Carbon never forms ionic C⁴⁺ or C⁴⁻ ions due to extreme energy barriers; it exclusively shares electron pairs, forming strong covalent bonds.

Step-by-Step Chart Preparation Method:-

Our team compiled and designed the classroom chart following these organized steps:

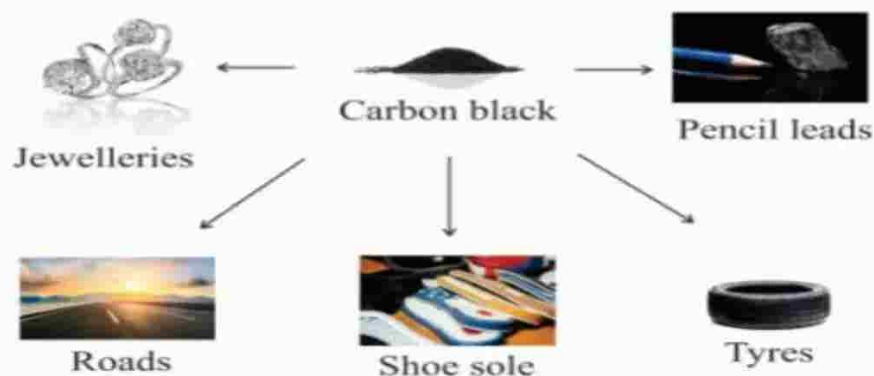
- **Step 1: Information Gathering:** Researched household items from our kitchen, medicine cabinet, bathroom, and school bags to identify their chemical organic compounds.
- **Step 2: Designing Grid Columns:** Ruled a 6-column matrix on our large white chart: Common Name, IUPAC Chemical Name, Molecular Formula, Structural Formula, Daily Life Use, and Family.
- **Step 3: Drawing Structural Formulas:** Drew clear chemical bond diagrams using black ink for carbon skeletons and blue ink for functional groups (-OH, -COOH, -CHO, -COO-).
- **Step 4: Color Coding Families:** Highlighted fuels in orange, medical alcohols in green, foods and acids in yellow, and cleaning soaps in blue for quick visual learning.
- **Step 5: Classroom Display & Presentation:** Mounted the finished chart on the main science display board and conducted a 20-minute presentation and Q&A session with classmates.

Everyday Carbon Compounds Chart Table:-

The comprehensive data table below presents the core information compiled in our chart:

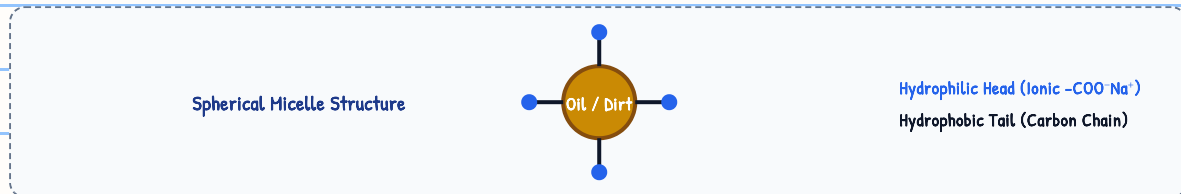
Everyday Item	IUPAC Name	Molecular Formula	Functional Group	Everyday Household Use
CNG / Bio-gas	Methane	CH_4	Alkane (C-C)	Clean domestic cooking fuel & green auto fuel
LPG Gas Cylinder	Butane + Propane	$\text{C}_4\text{H}_{10} + \text{C}_3\text{H}_8$	Alkane	LPG cooking gas in kitchen stoves
Hand Sanitizer	Ethanol	$\text{C}_2\text{H}_5\text{OH}$	Alcohol (-OH)	Antiseptic disinfectant; solvent in cough syrups
Cooking Vinegar	Ethanoic Acid	CH_3COOH	Carboxylic Acid (-COOH)	Preserving pickles and cooking Chinese food
Nail Polish Remover	Propanone (Acetone)	CH_3COCH_3	Ketone (>C=O)	Dissolving nail polish paint and laboratory solvent
Pineapple Essence	Ethyl Ethanoate	$\text{CH}_3\text{COOC}_2\text{H}_5$	Ester (-COO-)	Artificial flavoring in ice creams and perfumes

Uses of Carbon



Soaps & Micelle Action Analysis:-

Our chart included a detailed structural analysis of how soaps clean dirty clothes:



❖ *Cleansing Action of Soap (Micelle Chemistry):-*

- A soap molecule (like Sodium Stearate, $\text{C}_{17}\text{H}_{35}\text{COONa}$) has two distinct ends: a long non-polar hydrophobic hydrocarbon tail (water-repelling) and a polar hydrophilic ionic head ($\text{-COO}^- \text{Na}^+$, water-loving).
- When soap is dissolved in water, the hydrophobic carbon tails dissolve into the greasy oily dirt, while the ionic heads point outwards into the water.
- This forms a spherical cluster called a **Micelle**. Agitating the cloth lifts the trapped oil droplet into water, washing it away completely!

Classroom Seminar & Discussion Highlights:-

During the presentation of our chart in the science period, lively discussions took place:

- **Discussion 1: Why Carbon Forms Covalent Bonds Exclusively:** Students asked why carbon cannot simply gain 4 electrons to become C^{4-} or lose 4 electrons to become C^{4+} . We explained that a tiny carbon nucleus with 6 protons cannot hold 10 electrons (C^{4-}), nor can it surrender 4 electrons (C^{4+}) due to massive ionization enthalpy. Hence, sharing 4 electron pairs via covalent bonds is carbon's only stable path!
- **Discussion 2: Why Natural Vinegar Is Safe to Consume:** Vinegar is a 5% to 8% aqueous solution of ethanoic acid (acetic acid). Unlike strong mineral acids (HCl or sulfuric acid) that ionize completely in water, ethanoic acid is a weak organic acid that ionizes only partially, giving a pleasant tangy flavor without harming the stomach lining.
- **Discussion 3: The Menace of Synthetic Plastics:** Plastics like polythene (poly-ethene) and PVC are synthetic giant polymers that nature's bacteria cannot decompose, causing global plastic pollution.

Everyday Applications & Green Chemistry:-

Carbon chemistry impacts modern technology and sustainable environmental development:

- **1. Carbon Nanotubes & Graphene:** Allotropes of carbon like graphene (a single atomic layer of carbon) are 200 times stronger than steel and conduct electricity faster than copper, revolutionizing future touchscreens and electric car batteries!
- **2. Biofuels (Ethanol Blending in Petrol):** Blending 20% bio-ethanol (made from sugarcane juice fermentation) into vehicle petrol cuts carbon monoxide pollution and saves crude oil import costs.
- **3. Biodegradable Bio-Plastics (PLA):** Modern chemists are producing polylactic acid (PLA) plastics derived from corn starch and sugarcane that decompose naturally in garden compost in 90 days.

❖ Practical Household Takeaway:-

Replacing single-use plastic carry bags with reusable organic cotton or jute bags eliminates non-biodegradable synthetic polymers from our city drains!

What We Learned & Conclusion:-

- Carbon is a versatile tetravalent element that forms millions of compounds through unique catenation.
- Household items belong to distinct organic homologous series: fuels (alkanes), sanitizers (alcohols), vinegar (carboxylic acids), and perfumes (esters).
- Soaps clean grease by forming spherical micelles with hydrophobic carbon tails and hydrophilic ionic heads.
- Understanding carbon chemistry empowers us to adopt eco-friendly bio-plastics and clean bio-fuels.

Conclusion:-

Our project chart on carbon compounds in everyday use provided an eye-opening educational showcase for our entire class. From the fuels that cook our meals and the sanitizers that protect our health to the soaps that clean our clothes, carbon compounds form the very fabric of our daily existence. Preparing this chart demystified organic chemical formulas and inspired our classmates to appreciate the universal beauty of carbon chemistry.

References:-

- Maharashtra State Board Class 10 Science and Technology Part 1 Textbook (Topic: Carbon Compounds).
- NCERT Class 10 Science – Carbon and its Compounds.
- Everyday Organic Chemistry and Polymer Science Handbook, Balbharti Pune.